🧁 Team Muffin presents:

FirstSteps AI

Your Infant's First Voice. Decoded.

AI Nurse & Dermatologist for First-Time Parents
Powered by MedGemma · Kaggle MedGemma Impact Challenge 2025

Main Track | **Novel Task Track** | **MedGemma-1.5-4b-it**

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IT'S 3 AM.

Your newborn won't stop crying.

Priya and Arjun are first-time parents. Their 6-week-old has been crying for 40 minutes. Is she hungry? In pain? Is that red patch on her arm something serious?

They Google frantically. 15 browser tabs. Contradictory advice. Medical jargon. No real answers — just more panic.

Their pediatrician's office opens in 6 hours. And that strange patch on the baby's skin? They have no idea what it means.

Every first-time parent deserves a knowledgeable, calm voice at 3 AM.

FirstSteps-AI is that voice.

What is FirstSteps-AI?

Two AI-powered modes. One warm, expert voice. Always available.

🍼 MODE 1 — AI NURSE

1. Upload or record baby's cry audio
2. CNN0 confirms it's actually a cry
3. CNN1 ⟶ Hungry or Not-Hungry?
4. CNN2 ⟶ Belly pain / Tired / Burping / Discomfort
5. MedGemma delivers warm nurse guidance

🔬 MODE 2 — AI DERMATOLOGIST

1. Upload a photo of the baby's skin
2. CNN3 classifies from 7 conditions
3. Conditions: Eczema, Cellulitis, Ringworm, Urticaria, Scabies, Warts & more
4. MedGemma sees the image too
5. Gives 'When to Call Doctor' thresholds

Tone locked by prompt engineering: warm pediatric nurse talking to anxious first-time parents — never cold jargon.

Technical Architecture

Hierarchical CNN cascade → Mel spectrogram vision bridge → MedGemma clinical reasoning

🍼 CRY PIPELINE

AUDIO INPUT	CNN0 Cry Gate	CNN1 Hungry?	CNN2 Subtype	MedGemma 1.5-4B-IT
WAV/MP3/OGG	ResNet18 Kaggle-trained	ResNet18 DonateACry	ResNet18 4 subtypes	Warm nurse guidance

🔬 SKIN PIPELINE

IMAGE INPUT	CNN3 Skin Classify	7 Skin Conditions	MedGemma Visual Reason	Derm Guidance

Audio Bridge: Audio → Mel Spectrogram (128 mel bands, 22050 Hz) → 224×224 RGB PNG → ResNet18 | GPU: Google Colab T4

MedGemma as a Reasoning Engine

Novel Task: Multimodal fusion of audio waveform + acoustic features + CNN probabilities into a single clinical prompt

⚡ **KEY NOVELTY:** MedGemma is an **image-to-text / text-to-text** model — it has **no native audio understanding.** We made it reason about sound by converting cry audio into a Mel spectrogram image — feeding audio through MedGemma's ***vision encoder.***

Waveform Image

Cry audio rendered as a visual waveform — passed directly into MedGemma's vision encoder. Audio becomes a picture.

Acoustic Features

RMS, ZCR, MFCC coefficients, Spectral Centroid injected as clinical evidence in the text prompt

CNN Probabilities

Full softmax distribution from CNN1+2 shapes MedGemma's confidence language and advice tone

MedGemma Structured Output — 6 sections, nurse tone enforced by prompt engineering

- ✦ Condition Overview
- ✦ Immediate Actions
- ✦ Treatment & Relief
- ✦ Warning Signs
- ✦ **When to Call Doctor**
- ✦ **Warm Disclaimer**

Demo: Cry Analysis in Action

Priya records her baby's cry at 3 AM. Here's what happens in seconds.

01

Upload Audio

Priya uploads a 10-second WAV clip through the Gradio interface (WAV / MP3 / OGG / M4A)

02

CNN0 Gate

Model confirms it's a cry — not background noise or music. Cry probability: 0.91

03

CNN1 → Hungry?

Returns 'Not Hungry' — routes to CNN2 for subtype classification

04

CNN2 → Belly Pain (78%)

Subtype probabilities rendered as a horizontal bar chart alongside waveform, spectrogram & MFCC plots

05

MedGemma Reasoning

"Your baby's cry pattern suggests belly discomfort. Try the colic hold: face-down along your forearm..."

06

When to Call Doctor

'If crying continues past 3 hours, or temperature exceeds 38°C (100.4°F) — call your pediatrician now'

Demo: Skin Analysis in Action

Arjun notices a red patch on his baby's arm. He has no idea what it is.

7 CONDITIONS COVERED

- ✦ Atopic Dermatitis (Eczema)
- ✦ Cellulitis & Impetigo
- ✦ Drug Eruptions & Exanthems
- ✦ Tinea & Ringworm
- ✦ Scabies & Insect Bites
- ✦ Warts & Molluscum
- ✦ Urticaria (Hives)

What MedGemma Delivers

Condition Overview: What Atopic Dermatitis is, why it appears on infants, how common it is

What You Might See: Visual match description — red, scaly, dry patches often on cheeks or inner elbows

Immediate Care: Specific hygiene steps, clothing choices, and room humidity guidance for today

Safe Treatments: 1% hydrocortisone cream, fragrance-free emollients — named with usage instructions

When to Call Doctor: Spreading > 2cm in 24h? Fever > 38°C? Pus? Yellow crusting? Call immediately.

MedGemma receives both the CNN3 probability chart AND the original skin photo — true multimodal clinical reasoning.

Why FirstSteps-AI Wins Both Tracks

🏆 MAIN TRACK

Real Healthcare Impact

Helps 140M+ families/year make urgent, informed infant health decisions at home

MedGemma at the Core

4B-IT model used for structured multimodal clinical reasoning — not decoration

Responsible AI Design

Every response ends with a warm disclaimer; tool to assist, never replace pediatricians

Accessible 24/7

Free, instant guidance when clinics are closed and anxiety is at its highest

🚀 NOVEL TASK TRACK

Breaking MedGemma's Modality Limit

MedGemma is image/text only — no audio support. We made it reason about sound by converting cry waveforms into Mel spectrogram images, running audio through its vision encoder

Audio→Vision→LLM Bridge

Mel spectrograms convert raw audio into 224×224 visual representations, feeding a fundamentally non-audio model with acoustic intelligence it was never designed for

Triple-Signal Fusion Prompt

Waveform image (vision) + acoustic feature vectors + CNN softmax distribution — three signals fused into one structured clinical prompt

Pediatric Tone Engineering

Prompt engineering forces nurse-specific structured output with concrete medical thresholds — entirely new use of MedGemma's instruction-following capability

Datasets & Model Training

CNN0 — Cry Gate

Kaggle Infant Cry Dataset

→ Cry vs Non-cry binary classification

→ ResNet18 full fine-tune

→ Weighted random sampling for balance

→ 12 epochs · StepLR scheduler

CNN1 + CNN2 — Cry Type

DonateACry Corpus

→ 5 categories: hungry, belly_pain, burping, discomfort, tired

→ Audio → Mel Spectrogram → PNG

→ Hierarchical 2-stage pipeline

→ CNN2: layer4-only fine-tune (small dataset)

CNN3 — Skin Classifier

DermNet Dataset

→ 7 infant-relevant skin conditions

→ ResNet18 + class-weighted CrossEntropy

→ F1-macro as primary metric

→ 10 epochs · Adam · lr=1e-4

Pipeline: Audio → librosa Mel Spectrogram → 224×224 RGB PNG → ResNet18 CNN | Platform: Google Colab T4 GPU

Impact & Vision

140M+
babies born globally each year

3 AM
when parents need answers most

7
skin conditions at launch

5
cry types decoded with confidence

🌍 Global Reach
English is supported very well, and it explains everything in a simple manner

🏥 Clinical Integration
Triage partner for pediatric clinics flags high-risk cases for urgent escalation

📱 Mobile-First App
Record cry → snap photo → get guidance in under a minutes from any phone

🔬 Continuous Learning
Feedback loop with clinical outcomes to retrain CNNs and improve MedGemma prompts

FirstSteps-AI starts as a competition entry. It ends as infrastructure for infant healthcare.

🧁 Team Muffin

FirstSteps-AI

Because every first-time parent deserves a knowledgeable, calm voice — at any hour.

🍼 Cry decoded → CNN pipeline → MedGemma → Warm nurse guidance

🔬 Skin photo → CNN3 → MedGemma sees the image → Derm explanation

🏆 Main Track: Real healthcare impact for 140M+ families globally

🚀 Novel Task: Audio-to-LLM pipeline — first of its kind with MedGemma

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